

BIG DATA PROCESSING.

Lab02: On Large-Scale Transportation-Related Datasets.

BACKGROUND.

From the window you can see the sun shining in a lovely autumn morning. Its Monday, 10am, and you are in the open plan office of a new start-up, OptimiseYourJourney, which will enter the market next year with a clear goal in mind: “*leverage Big Data technologies for improving the user experience in transportation*”.

The start-up is at a very early stage, and has no clear product in mind yet. However, they have offered a short-term internship in their Big Data Engineering Department to help them exploring the datasets, technologies and techniques that can be applied in their future products. They do not pay very well (0€ per hour), but you see this as a good opportunity to complement your knowledge in the module Big Data Processing you are studying at the moment, so you have decided to give it a go.

OptimiseYourJourney



In the department meeting that has just finished your boss was particularly happy. During the weekend he was exploring the public datasets provided by the Irish government at <https://data.gov.ie/> and he found a transportation dataset named **Dublin Bus GPS sample data from Dublin City Council (Insight Project)**: <https://data.gov.ie/dataset/dublin-bus-gps-sample-data-from-dublin-city-council-insight-project> The original dataset contains 40+ millions of GPS data measurements collected from Dublin buses operating in January of 2013. Once extracted, it occupies ~5GB and is available at: <http://opendata.dublincity.ie/TrafficOpenData/sir010113-310113.zip>

Your boss thinks this dataset provides a great opportunity to explore the potential of Spark Core and Spark SQL in analysing large datasets. He has already cleaned the dataset for you to perform some data analysis on it.

DATASET.

The new dataset (obtained after cleaning the original one) is provided to you in the folder **Canvas => 5_Assignments => A01_dataset.zip**.

It occupies ~3GB and contains 744 files, one per hour interval and day of the month:

- siri.2013010100.csv => Provides the data measurements of 01st of January 2013 in the hour interval [12am, 1am)
- siri.2013010101.csv => Provides the data measurements of 01st of January 2013 in the hour interval [1am, 2am)
- ...
- siri.2013010108.csv => Provides the data measurements of 01st of January 2013 in the hour interval [8am, 9am)
- siri.2013010109.csv => Provides the data measurements of 01st of January 2013 in the hour interval [9am, 10am)
- ...
- siri.2013010123.csv => Provides the data measurements of 01st of January 2013 in the hour interval [11pm, 12am)
- siri.2013010200.csv => Provides the data measurements of 02nd of January 2013 in the hour interval [12am, 1am)
- ...
- siri.2013013123.csv => Provides the data measurements of 31st of January 2013 in the hour interval [11pm, 12am)

Each row of a file contains the following fields:

Date , Bus_Line , Bus_Line_Pattern , Congestion , Longitude , Latitude , Delay , Vehicle , Closer_Stop , At_Stop

- **(00) Date**
 - A String representing the date of the measurement with format <%Y-%m-%d %H:%M:%S>
 - Example: "2013-01-01 13:00:02"
- **(01) Bus_Line**
 - An Integer representing the bus line.
 - Example: 120
- **(02) Bus_Line_Pattern**
 - A String identifier for the sequence of stops scheduled in the bus line (different buses of the same bus line can follow different sequence of stops in different iterations).
 - Example: "027B1001" (it can also be empty "").
- **(03) Congestion**
 - An Integer representing whether the bus is at a traffic jam (No => 0 / Yes => 1) .
 - Example: 0
- **(04) Longitude**
 - A Float representing the longitude position of the bus.
 - Example: -6.269634
- **(05) Latitude**

- A Float representing the latitude position of the bus.
- Example: 53.360504
- **(06) Delay**
 - An integer representing the delay of the bus with respect to its schedule (measured in seconds). It is a negative value if the bus is ahead of schedule.
 - Example: 90.
- **(07) Vehicle**
 - An integer identifier for the bus vehicle.
 - Example: 33304.
- **(08) Closer_Stop**
 - An integer identifier for the closest bus stop.
 - Example: 7486.
- **(09) At_Stop_Stop**
 - An integer representing whether the bus vehicle is at the bus stop right now (i.e., stopping at it for passengers to hop on / hop off). (No -> 0 and Yes -> 1)
 - Example: 0.

TASKS / EXERCISES.

Both Assignment 1 and Assignment 2 will have as reference the aforementioned public transportation dataset **Dublin Bus GPS sample data from Dublin City Council (Insight Project)**.

However, there are many other publicly available transportation-related datasets out there (they can be focused on cars, buses, trains, taxis, bikes, scooters, etc).

Survey these alternative datasets, and select a dataset of interest to you:

- A publicly available transportation-related dataset (different from the Dublin Bus dataset).

Compare and contrast the new dataset w.r.t. the Dublin Bus dataset, including:

- A brief description of the dataset.
- Its main characteristics: URL for accessing to it, size, format of the data, etc.
- Its relevance - Think of potential use-cases derived from the dataset you are proposing that cannot be achieved with the Dublin Bus Dataset.